

Möbius inversion at the objective level

This is the last time I will be presenting at the SUSTech Graduate Topology Seminar during my master's studies. This note is a tribute to all the teachers and friends who have supported me along the way — we shared a wonderful time together, and I am grateful to you for witnessing my growth.

The convolution of the zeta function and the Möbius function returns to the origin, yet our lives move ever forward ☺

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This note is mainly written on a series of papers. For detailed research on decomposition spaces (used to construct incidence coalgebra in this note) and its application in combinatorics, one can read [2],[3],[4]; For the theory of Möbius inversion, one can refer to [5], and the most key point is summarised in the first section of this note; For the detailed version of cardinality (a tool transposes categorified theory to numbers or vector spaces), I recommend [1]; Besides, I also recommend a further reading about negative sets, which is a generalization of negative numbers, in [6].

1 Classical Möbius inversion

Construction 1.1. (Convolution algebra) Let (C, Δ, ε) be a coalgebra and (A, m, u) be an algebra. Denote $\Delta(x) = \sum_{(x)} x_{(1)} \otimes x_{(2)}$. Define the convolution product on $\text{Lin}(C, A)$ by $(\alpha * \beta)(x) = \sum_{(x)} m(\alpha(x_{(1)}), \beta(x_{(2)}))$. Then $(\text{Lin}(C, A), *, u\varepsilon)$ forms an algebra, called the convolution algebra. $\square$

Classical generalized Möbius inversion studies which elements of $\text{Lin}(C, A)$ are invertible for convolution.

Example 1.2. $(\mathbb{C}[\mathbb{N}^\times], \Delta, \varepsilon)$ is a coalgebra, where

$$\Delta(n) := \sum_{d|n} d \otimes \frac{n}{d}, \quad \varepsilon(n) = \begin{cases} 1, & n = 1, \\ 0, & \text{otherwise} \end{cases}$$

$\mathbb{C}$ is an algebra. Since a linear map is determined on the basis, we have

$$\text{Lin}(\mathbb{C}[\mathbb{N}^\times], \mathbb{C}) = \{f : \mathbb{N}^\times \rightarrow \mathbb{C}\}.$$

In the following we also represent $\zeta, \mu \in \text{Lin}(\mathbb{C}[\mathbb{N}^\times], \mathbb{C})$ by maps $\mathbb{N}^\times \rightarrow \mathbb{C}$.

Consider

$$\zeta : \mathbb{N}^\times \rightarrow \mathbb{C} \text{ defined by } n \mapsto 1$$

and

$$\mu : \mathbb{N}^\times \rightarrow \mathbb{C} \text{ defined by } n \mapsto \begin{cases} 0, & \text{if } n \text{ contains a square factor,} \\ (-1)^r, & \text{if } n \text{ is the product of } r \text{ distinct primes.} \end{cases}$$

Then one can check $\zeta * \mu = \mu * \zeta = \varepsilon$. This is the classical Möbius inversion principle. $\square$

Theorem 1.3. If $\phi \in \text{Lin}(C, A)$ sends all group-like elements ($\Delta(x) = x \otimes x$) to 1, then ϕ is convolution invertible. Its inverse ψ is given recursively by

$$\psi = e - \psi * (\phi - e), \quad e := u\varepsilon.$$

Proof sketch. We only show the construction of ψ . Let

$$\psi_0 = e, \quad \psi_{n+1} := \psi_n * (\phi - e).$$

Define

$$\psi_{\text{even}} := \sum_{n \text{ even}} \psi_n, \quad \psi_{\text{odd}} := \sum_{n \text{ odd}} \psi_n,$$

and then set

$$\psi := \psi_{\text{even}} - \psi_{\text{odd}}.$$

$\square$

In the next section we will formulate the above theorem at the objective level, i.e. categorify the classical generalized Möbius inversion theorem.

2 Categorified analogue of the coalgebra (C, Δ, ε)

Coalgebra is a vector space with compatible comultiplication.

Notation. E, B are groupoids. X is a simplicial space, i.e., a functor $\Delta^{op} \rightarrow \mathcal{S}$, where $\mathcal{S}$ is the ∞ -category of spaces. Write $\lceil b \rceil : 1 \rightarrow B$ for the map picking out b .

Definition 2.1. (Homotopy fiber) Given $f : E \rightarrow B$ and $b \in B$, define the homotopy fiber E_b by the homotopy pullback square

$$\begin{array}{ccc} E_b & \longrightarrow & E \\ \downarrow & \lrcorner & \downarrow f \\ 1 & \xrightarrow{\lceil b \rceil} & B. \end{array}$$

Fact 2.2. For any $f : E \rightarrow B$ in $\text{Grpd}_{/B}$, $f \simeq \int^{b \in B} E_b \lceil b \rceil$. $\square$

This suggests that $\text{Grpd}_{/B}$ behaves like a vector space:

- Basis: $\lceil b \rceil : 1 \rightarrow B \quad ([b] \in \pi_0 B)$
- Scalar multiplication: $A \lceil b \rceil$ For $A \in \mathbf{Grpd}$, define

$$A \lceil b \rceil := \left[A \rightarrow 1 \xrightarrow{\lceil b \rceil} B \right] \in \mathbf{Grpd}_{/B}.$$

- Addition: homotopy sum

□

Warning.

- 1) We have only defined “scalar multiplication” for $\lceil b \rceil : 1 \rightarrow B$, not arbitrary $E \rightarrow B$ in $\mathbf{Grpd}_{/B}$.
- 2) There is no negative sign here; for example $-\lceil b \rceil$ is meaningless.

□

Anyway, we obtain a categorified analogue of a vector space. Next we define comultiplication and counit.

We want to construct

$$\mathbf{Grpd}_{/B} \rightarrow \mathbf{Grpd}_{/B} \times \mathbf{Grpd}_{/B}, \quad \mathbf{Grpd}_{/B} \rightarrow \mathbf{Grpd}_{/1}.$$

Step 1. Using

$$\mathbf{Grpd}_{/B} \times \mathbf{Grpd}_{/B} \cong \mathbf{Grpd}_{/B \times B}, \quad \mathbf{Grpd}_{/1} \cong \mathbf{Grpd},$$

Step 2. A span $A \xleftarrow{f} C \xrightarrow{g} D$ induces a functor

$$\mathbf{Grpd}_{/A} \xrightarrow{f^*} \mathbf{Grpd}_{/C} \xrightarrow{g_*} \mathbf{Grpd}_{/D}.$$

$$\begin{array}{ccccccc} & & E' & \longrightarrow & E & & \\ & & \downarrow & & \downarrow & & \\ E \rightarrow A & \longmapsto & C & \xrightarrow{f} & A & \longmapsto & E' \rightarrow C \rightarrow D \end{array}$$

Remark 2.3. Since a classical coalgebra structure is bilinear, the functor $\mathbf{Grpd}_{/B} \rightarrow \mathbf{Grpd}_{/B \times B}$ should be represented by a span, which is the analogue of linear maps in category theory. □

Step 3. It is difficult to define a coalgebra structure on $\mathbf{Grpd}_{/B}$ for arbitrary $B \in \mathbf{Grpd}$. But the simplicial space will provide enough data for constructing. Let $X : \Delta^{\text{op}} \rightarrow \mathcal{S}$ be a simplicial space.

Remark 2.4. By definition, X is an ∞ -groupoid. But in many cases we only use simplicial groupoids, viewing an ordinary groupoid as an ∞ -groupoid. □

If $B = X_1$, then we have a span

$$X_1 \xleftarrow{d_1} X_2 \xrightarrow{(d_2, d_0)} X_1 \times X_1 \text{ which induces } \Delta : \mathbf{Grpd}_{/X_1} \rightarrow \mathbf{Grpd}_{/X_1 \times X_1}$$

and also a span

$$X_1 \xleftarrow{s_0} X_0 \rightarrow 1 \text{ which induces } \varepsilon : \mathbf{Grpd}_{/X_1} \rightarrow \mathbf{Grpd}_{/1}$$

Step 4. We require Δ and ε to be coassociative and counital. These properties hold when X is a *decomposition space*.

The coalgebra $(\mathrm{Grpd}_{/X_1}, \Delta, \varepsilon)$ is called the *incidence coalgebra* of the decomposition space X .

We take associativity for example: We require the following square of slice groupoids to commute:

$$\begin{array}{ccc} \mathrm{Grpd}_{/X_1} & \xrightarrow{\Delta} & \mathrm{Grpd}_{/X_1 \times X_1} \\ \Delta \downarrow & & \downarrow \Delta \otimes \mathrm{id} \\ \mathrm{Grpd}_{/X_1 \times X_1} & \xrightarrow{\mathrm{id} \otimes \Delta} & \mathrm{Grpd}_{/X_1 \times X_1 \times X_1} \end{array}$$

At the level of spans, one is tempted to look at

$$\begin{array}{ccccc} X_1 & \xleftarrow{d_1} & X_2 & \xrightarrow{(d_2, d_0)} & X_1 \times X_1 \\ d_1 \uparrow & & & & \uparrow d_1 \times \mathrm{id} \\ X_2 & & & & X_2 \times X_1 \\ (d_2, d_0) \downarrow & & & & \downarrow (d_2, d_0) \times \mathrm{id} \\ X_1 \times X_1 & \xleftarrow{\mathrm{id} \times d_1} & X_1 \times X_2 & \xrightarrow{\mathrm{id} \times (d_2, d_0)} & X_1 \times X_1 \times X_1 \end{array}$$

However, in this diagram, the notion of commutativity is vacuous.

Fact 2.5. For any homotopy pullback square

$$\begin{array}{ccc} H & \xrightarrow{q} & E \\ p \downarrow & \lrcorner & \downarrow g \\ G & \xrightarrow{f} & B, \end{array}$$

the functors

$$q_! p^*, g^* f_! : \mathrm{Grpd}_{/G} \rightarrow \mathrm{Grpd}_{/E}$$

are naturally homotopy equivalent. □

Instead, decompose the above diagram into four smaller squares:

$$\begin{array}{ccccc} X_1 & \xleftarrow{d_1} & X_2 & \xrightarrow{(d_2, d_0)} & X_1 \times X_1 \\ d_1 \uparrow & (1) & d_1 \uparrow & (2) & \uparrow d_1 \times \mathrm{id} \\ X_2 & \xleftarrow{d_2} & X_3 & \xrightarrow{(d_3, d_0 d_0)} & X_2 \times X_1 \\ (d_2, d_0) \downarrow & (3) & \downarrow (d_2 d_2, d_0) & (4) & \downarrow (d_2, d_0) \times \mathrm{id} \\ X_1 \times X_1 & \xleftarrow{\mathrm{id} \times d_1} & X_1 \times X_2 & \xrightarrow{\mathrm{id} \times (d_2, d_0)} & X_1 \times X_1 \times X_1 \end{array}$$

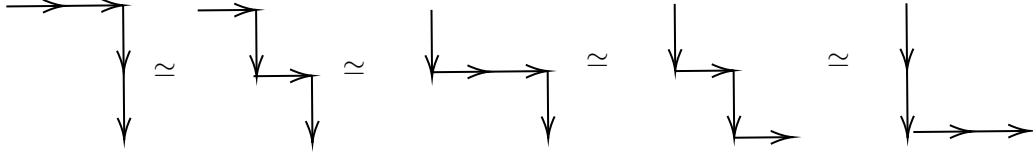
Squares (1) and (4) commute because simplicial object satisfying $d_i d_j = d_{j-1} d_i$ for all $i < j$. For example, square (4) commutes meaning that $(d_2 d_3 x, d_0 d_3 x, d_0 d_0 x) = (d_2 d_2 x, d_2 d_0 x, d_0 d_0 x)$ which is equivalent to $d_2 d_3 = d_2 d_2$ and $d_0 d_3 = d_2 d_0$.

Assume now that squares (2) and (3) are homotopy pullbacks. Then consider the morphism induced between slice groupoids.

$$\begin{array}{ccccc}
\mathrm{Grpd}_{/X_1} & \xrightarrow{d_1^*} & \mathrm{Grpd}_{/X_2} & \xrightarrow{(d_2, d_0)!} & \mathrm{Grpd}_{/X_1 \times X_1} \\
\downarrow & & \downarrow & & \downarrow \\
\mathrm{Grpd}_{/X_2} & \xrightarrow{\quad} & \mathrm{Grpd}_{/X_3} & \xrightarrow{\quad} & \mathrm{Grpd}_{/X_2 \times X_1} \\
\downarrow & & \downarrow & & \downarrow \\
\mathrm{Grpd}_{/X_1 \times X_1} & \xrightarrow{\quad} & \mathrm{Grpd}_{/X_1 \times X_2} & \xrightarrow{\quad} & \mathrm{Grpd}_{/X_1 \times X_1 \times X_1}
\end{array}$$

$(1)'$ $(2)'$ $(3)'$ $(4)'$

We have equivalences of paths in above diagram and this implies the original coassociativity square commutes.



The first and fourth equivalence of paths is because $(2)'$ and $(3)'$ are pullbacks, respectively. The second and third equivalence of paths is because $(1)'$ and $(4)'$ commute, respectively.

So the axioms of a decomposition space should imply that the (2) and (3) are pullbacks.

3 Categorical analogues of (A, m, u) and $\mathrm{Lin}(C, A)$

Construction 3.1. (Categorified algebra) $(\mathrm{Grpd}, \times, 1)$ is an algebra, where $X \times Y$ is the Cartesian product of groupoids. $\square$

Construction 3.2. (Categorified convolution algebra) Let $\mathbf{LIN}$ be the category of slice groupoids whose morphisms are induced by spans.

For $F, G \in \mathbf{LIN}(\mathrm{Grpd}_{/X_1}, \mathrm{Grpd})$, suppose F is induced by the span $X_1 \leftarrow M \rightarrow 1$ and G is induced by $X_1 \leftarrow N \rightarrow 1$. Define their convolution by the span

$$X_1 \leftarrow M * N \rightarrow 1,$$

which is defined as the composition of spans:

$$\begin{array}{ccccc}
& X_1 & & & \\
& \uparrow d_1 & \swarrow & & \\
& X_2 & \xleftarrow{\quad} & M * N & \\
& \downarrow (d_2, d_0) & \searrow & \downarrow & \searrow \\
X_1 \times X_1 & \xleftarrow{\quad} & M \times N & \xrightarrow{\quad} & 1
\end{array}$$

The neutral functor for convolution is $\varepsilon : \mathrm{Grpd}_{/X_1} \rightarrow \mathrm{Grpd}$. the same ε function in the incidence coalgebra.

$(\mathbf{LIN}(\mathrm{Grpd}_{/X_1}, \mathrm{Grpd}), *, \varepsilon)$ is called the convolution algebra. $\square$

Fact 3.3. $\mathrm{Grpd}_B \simeq \mathrm{Grpd}^B$. In particular, the convolution algebra

$$\mathbf{LIN}(\mathrm{Grpd}_{/X_1}, \mathrm{Grpd}) \simeq \mathrm{Grpd}^{X_1}.$$

$\square$

4 Cardinality

Taking cardinality turns the objective-level theory into the classical one.

Definition 4.1. If X is a finite groupoid, define the cardinality of X be $|X| := \sum_{[x] \in \pi_0 X} \frac{1}{|\text{Aut}(x)|} \in \mathbb{Q}$.

Fact 4.2. Let A, B, G be locally finite groupoids and let $A \leftarrow G \rightarrow B$ be a finite span. Then the induced functor $\text{Grpd}_{/A} \rightarrow \text{Grpd}_{/B}$ restricts to $\text{grpd}_{/A} \rightarrow \text{grpd}_{/B}$.

Construction 4.3. Let $\mathbf{lin}$ be the category of slice categories $\text{grpd}_{/A}$, with A locally finite. There is a functor

$$\| \cdot \| : \mathbf{lin} \rightarrow \mathbf{Vect}$$

$$\begin{array}{ccc} \text{grpd}_{/A} & \xrightarrow{\quad} & \mathbb{Q}_{\pi_0 A} \\ \downarrow & & \downarrow \\ \text{grpd}_{/B} & \xrightarrow{\quad} & \mathbb{Q}_{\pi_0 B} \end{array} \quad \begin{array}{c} \delta_a \\ \downarrow \\ \sum_{[b] \in \pi_0 B} |B_{[b]}| |G_{a,b}| \delta_b \end{array}$$

Here

- $\mathbb{Q}_{\pi_0 A} := \mathbb{Q}[\pi_0 A]$ is the vector space with basis $\{\delta_a\}_{[a] \in \pi_0 A}$
- $B_{[b]}$ is the full subgroupoid of B on objects isomorphic to b
- $G_{a,b}$ is the fiber of $G \rightarrow A \times B$ over (a, b)

□

Construction 4.4. Dually there is a functor

$$\| \cdot \| : \text{grpd}^B \rightarrow \text{grpd}^A$$

$$\begin{array}{ccc} \text{grpd}^B & \xrightarrow{\quad} & \mathbb{Q}^{\pi_0 B} \\ \downarrow & & \downarrow \\ \text{grpd}^A & \xrightarrow{\quad} & \mathbb{Q}^{\pi_0 A} \end{array} \quad \begin{array}{c} \delta^b \\ \downarrow \\ \sum_{[a] \in \pi_0 A} |B_{[b]}| |G_{a,b}| \delta^a \end{array}$$

Here $\mathbb{Q}^{\pi_0 B}$ is the function space, i.e. the profinite-dimensional vector space with pro-basis given by the characteristic functions δ^b . □

Remark 4.5. One can think of

$$\mathbb{Q}^{\pi_0 B} = \varprojlim_{J \subset \pi_0 B, J \text{ finite}} \mathbb{Q}^J,$$

each $\mathbb{Q}^J$ finite dimensional. So a vector in $\mathbb{Q}^{\pi_0 B}$ can be represented by combination of probasis with infinite terms of nonzero coefficients. □

5 Möbius inversion at the objective level

5.1 Categorized zeta function

We want to define $\zeta : \text{Grpd}_{/X_1} \rightarrow \text{Grpd}$. Suppose ζ is represented by a span $X_1 \xleftarrow{f} G \rightarrow 1$.

Classically, ζ is the constant function 1, we hope after taking cardinality, ζ becomes

$$\mathbb{Q}_{\pi_0 X_1} \rightarrow \mathbb{Q}, \quad \delta_x \mapsto 1.$$

Now let's find what ζ should be. Suppose ζ is given by $X_1 \xleftarrow{f} G \rightarrow 1$.

$$|\zeta| : \mathbb{Q}_{\pi_0 X_1} \rightarrow \mathbb{Q},$$

$$\delta_x \mapsto \sum_{[b] \in \pi_0 1} |1_{[b]}| |G_{x,b}| \delta_b = |G_{x,*}| \delta_*$$

Remark 5.1. $1_{[*]}$ is the full subcategory of 1 consisting of objects isomorphic to $*$. So $|1_{[*]}| = 1$ $\square$

Hence we need to construct G and f , such that $|G_{x,*}| = 1$, for all $[x] \in \pi_0 X_1$.

If you thought of $G \rightarrow X_1 \times 1$ as morphism between sets and $|G_{x,*}|$ be the cardinality of the preimage set of $(x, *)$, the cheapest map to realize this is $X_1 \xleftarrow{\text{id}} X_1 \rightarrow 1$. We now check $X_1 \xleftarrow{\text{id}} X_1 \rightarrow 1$ will meet our requirement.

$G_{x,*}$ is the pullback

$$\begin{array}{ccc} G_{x,*} & \longrightarrow & X_1 \\ \downarrow & \lrcorner & \downarrow \\ 1 & \xrightarrow{(x,*)} & X_1 \times 1. \end{array}$$

For two objects $(*, y, \phi : x \rightarrow y)$, $(*, y', \phi' : x \rightarrow y') \in G_{x,*}$, there is a morphism

$$(\text{id}, \phi' \phi^{-1}) : (*, y, \phi) \rightarrow (*, y', \phi')$$

between them, so $\pi_0 G_{x,*}$ is a singleton.

Remark 5.2. $(\text{id}, \phi' \phi^{-1})$ is a morphism, since the following diagram commutes.

$$\begin{array}{ccc} (x, *) & \xrightarrow{\text{id}} & (x, *) \\ \phi \downarrow & & \downarrow \phi' \\ (y, *) & \xrightarrow{\phi' \phi^{-1}} & (y', *) \end{array}$$

$\square$

Let $(*, y, \phi) \xrightarrow{(\text{id}, g)} (*, y, \phi)$ be an automorphism, then the above commutative diagram forces $g = \text{id}_y$. Therefore $|\text{Aut}(*, y, \phi)| = 1$. Hence $|G_{x,*}| = \frac{1}{|\text{Aut}(*, y, \phi)|} = 1$, which is precisely the desired result.

So the objective-level zeta functor is represented by $X_1 \xleftarrow{\text{id}} X_1 \rightarrow 1$.

5.2 Inverse of ζ at the objective level

Idea and problem: In the classical case, we construct a family of $\{\phi_n\}$ and let

$$\mu = \sum_{n \text{ even}} \phi_n - \sum_{n \text{ odd}} \phi_n.$$

Then can check $\zeta * \mu = \epsilon$. Note that in our case, ϕ_{odd} lies in $\text{Grpd}^{/X_1}$, where the “ $-\phi_{\text{odd}}$ ” does not make sense. So the right formula suggests that the Möbius inversion principle should be

$$\zeta * \sum_{n \text{ even}} \phi_n = \epsilon + \zeta * \sum_{n \text{ odd}} \phi_n,$$

where the negative sign disappears.

Before constructing Φ_n , we first introduce completeness, which is necessary for defining incompletion of groupoids.

5.3 Completeness

Definition 5.3. A decomposition space X is *complete* if $s_0 : X_0 \rightarrow X_1$ is monomorphism, i.e., each homotopy fiber of s_0 is either empty or contractible. $\square$

Example 5.4. Let G be a nontrivial group, and let X be the simplicial groupoid with $X_n = (BG)^n$. Then $s_0 : 1 \rightarrow BG$ is not monomorphism, so X is not complete.

Indeed, there is only one object in BG and the only fiber groupoid is

$$\begin{cases} \text{objects:} & (*, *, g \in G) \\ \text{morphisms:} & \text{Hom}((*, *, g_1), (*, *, g_2)) = \begin{cases} (\text{id}, \text{id}) & g_1 = g_2 \\ \emptyset & \text{o/w} \end{cases} \end{cases}$$

This fiber groupoid is discrete but not connected, so it's not contractible. Hence, s_0 is not a monomorphism.

Another argument: Suppose on the contrary that s_0 is a monomorphism, then s_0 would be fully faithful, so $s_0(1)$ would be the full subgroupoid of BG . That would force $\text{id} = \text{Hom}_1(*, *) \simeq \text{Hom}_{BG}(*, *) = G$, impossible when G is nontrivial. $\square$

Motivation for completeness

We want to define nondegenerate simplices for a simplicial groupoid X .

For simplicial sets, the set of nondegenerate simplices is the complement of the image of degeneracy maps. For simplicial groupoids, one would like the complement of the image *groupoid*. But in general that complement need not itself be a groupoid.

Example 5.5. Let $\mathcal{C}$ be a category with one object and only one nontrivial morphism e with $e^2 = \text{id}$. Consider $1 \rightarrow \mathcal{C}$. The complement is a morphism e , does not contain any objects. So the complement is not a groupoid. $\square$

So to make the complement of the image meaningful, we want the degeneracy map to be fully faithful, i.e. a monomorphism.

Fact 5.6. In a decomposition space, a simplex is nondegenerate if and only if all its principal edges are nondegenerate. $\square$

Example 5.7. (Principal edges) In a 2-simplex the principal edges are x_{01} and x_{12} , not x_{02} . In a 3-simplex the principal edges are x_{01}, x_{12}, x_{23} .

Hence, it suffices to make nondegenerate 1-simplices make sense, i.e. it suffices to make the complement of $s_0 : X_0 \rightarrow X_1$ makes sense. This illustrates the condition of completeness.

Constructing Φ_n

Classical situation

$$\Phi_0 = \varepsilon, \quad \Phi_1 = \zeta - \varepsilon, \quad \Phi_n = \Phi_{n-1} * (\zeta - \varepsilon).$$

Then $\Phi_n = (\zeta - \varepsilon)^n = \Phi_1^n$.

Objective level

Define $\vec{X}_n \subset X_n$ to be the full subgroupoid of nondegenerate n -simplices. ε is induced by span Φ_0 is induced by $X_1 \xleftarrow{s_0} X_0 \rightarrow 1$, and ζ is induced by span $X_1 \xleftarrow{\text{id}} X_1 \rightarrow 1$. Hence it's reasonable to define $\Phi_1 = \zeta - \varepsilon$ induced by $X_1 \leftarrow \vec{X}_1 \rightarrow 1$. More generally, define Φ_n induced by span $X_1 \leftarrow \vec{X}_n \rightarrow 1$ where $\vec{X}_0 = X_0$ and $\vec{X}_n \rightarrow X_1$ is the restriction of the canonical morphism $X_n \rightarrow X_1$.

The next proposition illustrate our Φ_n have same behavior as classical case.

Remark 5.8. There is a canonical map $X_n \rightarrow X_1$ obtained by composing all arrows:

$$X_n \xrightarrow{d_{i_n}} X_{n-1} \xrightarrow{d_{i_{n-1}}} X_{n-2} \rightarrow \cdots \rightarrow X_2 \xrightarrow{d_{i_2}} X_1,$$

where each d_{i_j} is an inner face map (since we only consider composition of arrows), so $1 \leq i_j \leq j-1$.

Claim: $d_{i_2}d_{i_3} \cdots d_{i_n} = d_1d_1 \cdots d_1$.

Indeed, one repeatedly uses

$$d_{j-1}d_i = d_id_j$$

to move d_j leftward until its index drops to 1. This can always be achieved. There are $j-2$ maps preceding d_{i_j} . Since $1 \leq i_j \leq j-1$, the maximum value i_j can take is $j-1$. By commuting d_{i_j} to the left (moving it forward), one can ultimately shift it past $j-2$ maps to reduce the index by $j-2$; that is, $d_{j-1-(j-2)} = d_1$. For example,

$$X_5 \rightarrow X_1, \quad d_2d_1d_3d_4 = d_1d_2d_1d_3 = d_1d_1d_2d_1 = d_1d_1d_1d_1.$$

□

Property 5.9. $\Phi_n = (\Phi_1)^n$

Proof sketch. We prove by induction. Assume $\Phi_{n-1} = (\Phi_1)^{n-1}$. Then $(\Phi_1)^n = \Phi_{n-1} * \Phi_1$, which is given by

$$\begin{array}{ccccc} & X_1 & & & \\ & \uparrow & & & \\ & X_2 & \xleftarrow{\quad} & ? & \\ & \downarrow & & \downarrow & \\ X_1 \times X_1 & \xleftarrow{\quad} & \vec{X}_{n-1} \times \vec{X}_1 & \longrightarrow & 1 \end{array}$$

Using [Lemma 3.5, [3]] one can show $? = X_{a \dots a} = \vec{X}_n$, which shows $(\Phi_1)^n = \Phi_n$.

6 Sign-free Möbius inversion for complete decomposition spaces

Lemma 6.1. The square

$$\begin{array}{ccc} \vec{X}_1 + \vec{X}_2 & \longrightarrow & X_2 \\ \downarrow & \lrcorner & \downarrow (d_2, d_0) \\ X_1 \times \vec{X}_1 & \longrightarrow & X_1 \times X_1 \end{array}$$

is a pullback.

Proof. (While not strictly formal, this provides an intuitive sense of the underlying mechanism.) An object in the pullback is a tuple (a, b, σ, ϕ) with

$$(a, b) \in X_1 \times \vec{X}_1, \quad \sigma \in X_2, \quad \text{and } (a, b) \xrightarrow{\phi} (d_2\sigma, d_0\sigma).$$

Case 1. If a is degenerate, then σ is determined by $b \in \vec{X}_1$ (up to equivalence)

$$\begin{array}{ccc} & \bullet & \\ \text{id} \simeq d_2\sigma \nearrow & & \searrow d_0\sigma \simeq b \\ \bullet & \xrightarrow{d_1\sigma} & \bullet \end{array}$$

in this case $d_1\sigma \simeq b$. So (a, b, σ, ϕ) can be encoded in $b \in \vec{X}_1$ (up to equivalence)

Case 2. If a is nondegenerate, then (a, b, σ, ϕ) is encoded by a nondegenerate 2-simplex $\sigma \in \vec{X}_2$, whose principal edges are $d_2\sigma \simeq a$, $d_0\sigma \simeq b$, both are nondegenerate. For a complete decomposition space, nondegeneracy of the principal edges implies nondegeneracy of the whole simplex. Hence, $\sigma \in \vec{X}_2$. And in this case, $d_1\sigma \simeq b \circ a$. $\square$

Therefore,

$$\begin{array}{ccccc} & d_1\sigma & & X_1 & \\ & \uparrow & & \uparrow & \\ & \sigma & & X_2 & \longleftarrow \vec{X}_1 + \vec{X}_1 \\ & \downarrow & & \downarrow & \downarrow \\ (d_2\sigma, d_0\sigma) & & X_1 \times X_1 & \longleftarrow X_1 \times \vec{X}_1 & \longrightarrow 1 \\ & \searrow \phi \simeq & & & \\ & (a, b) & \longleftarrow & (a, b) & \end{array}$$

The above diagram shows

Corollary 6.2. $\zeta * \Phi_1 \simeq \Phi_1 + \Phi_2$. Higher-dimensional analogues gives $\zeta * \Phi_n \simeq \Phi_n + \Phi_{n+1}$. $\square$

Remark 6.3. For a rigorous version, there is a proof using “word notation”. See **Section 2.8**, **Lemma 2.12**, **Proposition 2.13**, **Proposition 3.7** in [3]. $\square$

Theorem. For a complete decomposition space,

$$\zeta * \Phi_{\text{even}} = \varepsilon + \zeta * \Phi_{\text{odd}}, \quad \Phi_{\text{even}} * \zeta = \varepsilon + \Phi_{\text{odd}} * \zeta,$$

where

$$\Phi_{\text{even}} := \sum_{k \geq 0} \Phi_{2k}, \quad \Phi_{\text{odd}} := \sum_{k \geq 0} \Phi_{2k+1}.$$

Proof.

$$\begin{aligned}\zeta * \Phi_{\text{even}} &= \zeta * \sum_{k \geq 0} \Phi_{2k} = \sum_{k \geq 0} \zeta * \Phi_{2k} = \sum_{k \geq 0} (\Phi_{2k} + \Phi_{2k+1}) = \sum_{k \geq 0} \Phi_k, \\ \zeta * \Phi_{\text{odd}} + \varepsilon &= \zeta * \left(\sum_{k \geq 0} \Phi_{2k+1} \right) + \Phi_0 = \sum_{k \geq 0} (\Phi_{2k+1} + \Phi_{2k+2}) + \Phi_0 = \sum_{k \geq 0} \Phi_k.\end{aligned}$$

Hence, $\zeta * \Phi_{\text{even}} = \zeta * \Phi_{\text{odd}} + \varepsilon$. Similarly, $\Phi_{\text{even}} * \zeta = \Phi_{\text{odd}} * \zeta + \varepsilon$.

The Möbius inversion will turn to classical one after taking cardinality.

Fact 6.4. If X is a Möbius decomposition space, then $|\zeta| * |\mu| = |\varepsilon| = |\mu| * |\zeta|$, where $|\mu| := |\Phi_{\text{even}}| - |\Phi_{\text{odd}}|$. $\square$

7 Example for $\mathbf{N} := N((\mathbb{N}, +))$

$(\mathbb{N}, +)$ is a monoid and $B(\mathbb{N}, +)$ is the category with one object with hom set be $(\mathbb{N}, +)$.

Let $\mathbf{N}$ be the nerve of the monoid $(\mathbb{N}, +)$.

7.1 Möbius function and its cardinality

The objective-level Φ_r . Φ_r is represented by the span $\mathbf{N}_1 \leftarrow \vec{\mathbf{N}}_r \rightarrow 1$. Here $\mathbf{N}_r = N(B\mathbb{N}) = \{(n_1, \dots, n_r) \mid n_i \in \mathbb{N}\}$, and the nondegenerate part $\vec{\mathbf{N}}_r = \{(n_1, \dots, n_r) \mid n_i \in \mathbb{N} \setminus \{0\}\} = (\mathbb{N} \setminus \{0\})^r$.

The fiber of

$$\vec{\mathbf{N}}_r \rightarrow \mathbf{N}_1 = \mathbb{N}, \quad (n_1, \dots, n_r) \mapsto n_1 + \dots + n_r$$

over $n \in \mathbb{N}$ is

$$(\vec{\mathbf{N}}_r)_n = \{(n_1, \dots, n_r) \in (\mathbb{N} \setminus \{0\})^r \mid n_1 + \dots + n_r = n\} = \{n \twoheadrightarrow r\},$$

the set of r -partitions of n .

Therefore, $\Phi_r(\ulcorner n \urcorner) = \{n \twoheadrightarrow r\}$.

Hence, $\Phi_{\text{even}}(n) = \{n \twoheadrightarrow r \mid r \text{ even}\}$, $\Phi_{\text{odd}}(n) = \{n \twoheadrightarrow r \mid r \text{ odd}\}$.

With an abusive sign notation one would like to write

$$\mu(n) = \sum_{r \geq 0} (-1)^r \{n \twoheadrightarrow r\}.$$

Remark 7.1. Note that here we use “negative sets”, which does not make sense. A discussion of negative sets can be found in [6].

Cardinality computation.

$$|\{n \twoheadrightarrow r\}| = \binom{n-1}{r-1}.$$

Remark 7.2. Since each cell must contain at least one element, the problem reduces to choosing $r-1$ positions for the bars from the $n-1$ available gaps between elements. $\square$

Thus $|\mu(n)| = \sum_{r \geq 0} (-1)^r \binom{n-1}{r-1}$. Writing $m := n-1$, $s := r-1$, this becomes $|\mu(n)| = \sum_{s \geq -1} (-1)^{s+1} \binom{m}{s}$.

- For $n = 1$ one gets $\mu(1) = (-1)^0 \binom{0}{-1} + (-1)^1 \binom{0}{0} = -1$,

- For $n = 0$ one gets $|\mu(0)| = \underbrace{(-1)^0 \binom{-1}{-1}}_1 + \underbrace{\binom{-1}{0} + \binom{-1}{1} + \dots}_0 = 1$
- For $n > 1$, $|\mu(n)| = \sum_{s=0}^m (-1)^{s+1} \binom{m}{s} = -(x-1)^m \big|_{x=1} = 0$.

Hence $|\mu(n)| = \begin{cases} 1, & n = 0, \\ -1, & n = 1, \\ 0, & \text{otherwise.} \end{cases}$

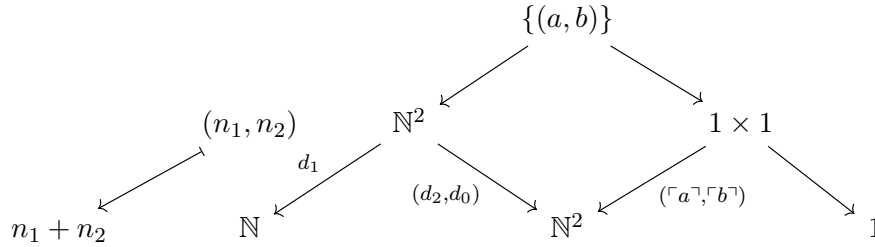
So, in $\mathbb{Q}^{\pi_0 \mathbb{N}} \cong \mathbb{Q}^{\mathbb{N}}$, $\mu = \sum |\mu(n)| \delta^n = \delta^0 - \delta^1$.

Another way to compute $\mu = \delta^0 - \delta^1$

Property 7.3. The incidence algebra $(\text{grpd}^{\mathbb{N}}, *, \varepsilon)$ has basis $\{h^n\}_{n \in \mathbb{N}}$ with convolution $h^a * h^b = h^{a+b}$.

Proof. Fact: grpd^S is the full subcategory of $\text{Grpd}^S \simeq \mathbf{LIN}(\text{Grpd}_S, \text{Grpd})$. So an object in grpd^S is a linear functor $\text{Grpd}_S \rightarrow \text{Grpd}$. Since functor is linear, it's induced by span $S \leftarrow G \rightarrow 1$. The basis $h^s \in \text{grpd}^S$ is induced by $S \xleftarrow{\lceil s \rceil} 1 \rightarrow 1$.

By the following diagram, one can show $h^a * h^b$ is given by $\mathbb{N} \leftarrow \{(a, b)\} \rightarrow 1$, $a + b \leftarrow (a, b)$, which is $\mathbb{N} \xleftarrow{\lceil a+b \rceil} 1 \rightarrow 1$. So $h^a * h^b = h^{a+b}$.



The cardinality of the incidence algebra is $\mathbb{Q}^{\pi_0 \mathbb{N}} = \mathbb{Q}^{\mathbb{N}}$.

Property 7.4.

$$\mathbb{Q}^{\mathbb{N}} \cong \mathbb{Q}[[z]], \quad \delta^n = |h^n| \longleftrightarrow z^n.$$

Proof.

$$f : \mathbb{N} \rightarrow \mathbb{Q} \quad \longleftrightarrow \quad \sum_{n \geq 0} f(n) z^n.$$

Here $f : \mathbb{N} \rightarrow \mathbb{Q}$ represents a vector in $\mathbb{Q}^{\mathbb{N}}$ whose coefficients of δ^n is $f(n)$. $|h^a * h^b| = |h^{a+b}|$, i.e., $\delta^a * \delta^b = \delta^{a+b}$

corresponds to $z^a \cdot z^b = z^{a+b}$.

In $\mathbb{Q}[[z]]$, $\zeta = \sum_{n \geq 0} z^n = \frac{1}{1-z}$, since $f(n) \equiv 1$. so its inverse is $\mu = 1 - z = 1 \cdot z^0 + (-1) \cdot z^1$, corresponding to

$$\mu = 1 \cdot \delta^0 + (-1) \cdot \delta^1 = \delta^0 - \delta^1 \text{ in } \mathbb{Q}^{\mathbb{N}}$$

7.2 Cancellation of $\Phi_{\text{odd}}(n)$ and $\Phi_{\text{even}}(n)$ for $n \geq 2$

At the objective level, the vanishing for $n \geq 2$ comes from an actual cancellation of groupoids. There is a bijection between odd and even decompositions.

We denote an r -partition of n elements in $\Phi_r(n)$ as $(x_1, \dots, x_r)$ for $x_i \in \mathbb{N}^*$ and $\sum_i x_i = n$.

For $n \geq 2$, there is a bijection

$$\Phi_{\text{odd}}(n) \longrightarrow \Phi_{\text{even}}(n)$$

by

$$(x_1, \dots, x_r) \longmapsto \begin{cases} (1, x_1 - 1, x_2, \dots, x_r), & \text{if } x_1 > 1, \\ (x_2 + 1, x_3, \dots, x_r), & \text{if } x_1 = 1. \end{cases}$$

Remark 7.5. This morphism is only well-defined for $n \geq 2$, since $\Phi_{\text{even}}(1) = \emptyset$. $\square$

Therefore, $|\Phi_{\text{even}}(n)| = |\Phi_{\text{odd}}(n)|$, $n \geq 2$. This would illustrate why $\mu(n) = 0$ for $n \geq 2$ from the objective level.

The fact that the Möbius function vanishes for $n \geq 2$ is not a coincidence of alternating sums; rather, at the objective level, the collections of odd and even decompositions for all $n \geq 2$ are themselves equivalent, which leads to their cancellation.

7.3 Check the Möbius principle

Goal: check $\zeta * \Phi_{\text{even}} = \zeta * \Phi_{\text{odd}} + \varepsilon$.

The counit ε is represented by $\mathbf{N}_1 \xleftarrow{s_0} \mathbf{N}_0 \rightarrow 1$, $0 \leftarrow *$, since $\text{id}_* = 0 \in \mathbb{N}$ is the identity under composition in $B\mathbb{N}$.

This induces $\epsilon : \text{Grpd}/\mathbf{N}_1 \longrightarrow \text{Grpd}/\mathbf{N}_0 \longrightarrow \text{Grpd}/_1 \simeq \text{Grpd}$

$$\begin{array}{ccccccc} & & ? & \longrightarrow & 1 & & \\ & & \downarrow & & \downarrow \lceil n \rceil & & \\ 1 & \xrightarrow{\lceil n \rceil} & \mathbb{N} & \longmapsto & * & \longrightarrow & \mathbb{N} \longmapsto ? = \begin{cases} \emptyset, & \text{when } n \neq 0 \\ *, & \text{when } n = 0 \end{cases} \end{array}$$

$$\text{Hence, } \varepsilon(\lceil n \rceil) = \begin{cases} \emptyset, & n \neq 0, \\ 1, & n = 0. \end{cases}$$

For $n \geq 2$, since $\Phi_{\text{even}}(n) \simeq \Phi_{\text{odd}}(n)$ and $\varepsilon(\lceil n \rceil) = \emptyset$, the identity

$$\zeta * \Phi_{\text{even}} = \zeta * \Phi_{\text{odd}} + \varepsilon$$

holds automatically. Then it remains to check the cases $n \leq 1$.

For $n \leq 1$ one has $\Phi_{\text{even}}(n) = \Phi_0(n) = \epsilon(n)$, $\Phi_{\text{odd}}(n) = \Phi_1(n)$. Therefore it suffices to show $\zeta * \epsilon = \zeta * \Phi_1 + \varepsilon$.

Recall:

- Φ_r is given by $\mathbb{N} \leftarrow (\mathbb{N} \setminus \{0\})^r \rightarrow 1$, $x_1 + \dots + x_r \leftarrow (x_1, \dots, x_r)$,
- ϵ is given by $\mathbb{N} \leftarrow * \rightarrow 1$, $0 \leftarrow *$,
- Φ_1 is given by $\mathbb{N} \leftarrow \mathbb{N} \setminus \{0\} \rightarrow 1$, $n \leftarrow n$,
- ζ is given by $\mathbb{N} \xleftarrow{\text{id}} \mathbb{N} \rightarrow 1$.

Property 7.6. For $n \leq 1$, $\Phi_1(\lceil n \rceil)$ coincides with $\delta^1(\lceil n \rceil)$ of the element 1

Proof. $\Phi_1 : \text{Grpd}_{/\mathbb{N}} \rightarrow \text{Grpd}$ is computed by

$$\begin{array}{ccccccc}
 & & ? & \longrightarrow & 1 & & \\
 & & \downarrow & & \downarrow \lceil n \rceil & & \\
 1 & \xrightarrow{\lceil n \rceil} & \mathbb{N} & \longmapsto & \mathbb{N} \setminus 0 & \longrightarrow & \mathbb{N} \longmapsto ? = \begin{cases} \emptyset, & n = 0 \\ \{n\}, & \text{otherwise} \end{cases}
 \end{array}$$

$\delta^1 : \text{Grpd}_{/\mathbb{N}} \rightarrow \text{Grpd}$ is computed by:

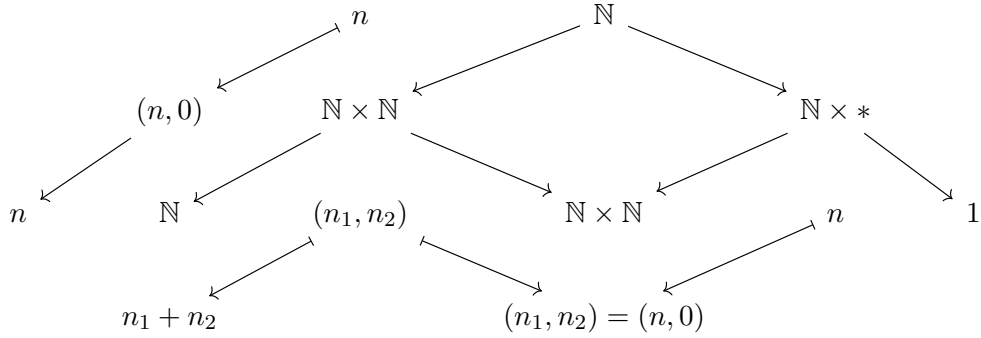
$$\begin{array}{ccccccc}
 & & ? & \longrightarrow & 1 & & \\
 & & \downarrow & & \downarrow \lceil n \rceil & & \\
 1 & \xrightarrow{\lceil n \rceil} & \mathbb{N} & \longmapsto & * & \longrightarrow & \mathbb{N} \longmapsto ? = \begin{cases} 1, & n = 1 \\ \emptyset, & \text{otherwise} \end{cases}
 \end{array}$$

Hence, $\Phi_1(\lceil n \rceil) = \delta^1(\lceil n \rceil)$ for $n \leq 1$. □

So it suffices to check $\zeta * \epsilon = \zeta * \delta^1 + \epsilon$ for $n \leq 1$.

Property 7.7. $\zeta * \epsilon$ is given by $\mathbb{N} \xleftarrow{\text{id}} \mathbb{N} \rightarrow 1$.

Proof.



□

Property 7.8. $\zeta * \delta^1$ is given by $\mathbb{N} \leftarrow \mathbb{N} \rightarrow 1$, $n + 1 \leftarrow n$,

Proof.

$$\begin{array}{ccccc}
 n_1 + n_2 & & \mathbb{N} & & \\
 \uparrow & & \uparrow & & \\
 (n_1, n_2) & & \mathbb{N} \times \mathbb{N} & \longleftarrow & \mathbb{N} \\
 \downarrow & & \downarrow & & \downarrow \\
 (n_1, n_2) & & \mathbb{N} \times \mathbb{N} & \xleftarrow{\text{id} \times \{1\}} & \mathbb{N} \times * \longrightarrow 1
 \end{array}$$

$(n, 1) \longleftarrow n$

□

Corollary 7.9. $(\zeta * \delta^1)(\lceil n \rceil) = \begin{cases} \{n-1\}, & n \geq 1, \\ \emptyset, & n = 0. \end{cases}$

Proof. $\zeta * \delta^1 : \text{Grpd}/\mathbf{N} \rightarrow \text{Grpd}$ computed by

$$\begin{array}{ccccccc}
 & & ? & \longrightarrow & 1 & & \\
 & & \downarrow & & \downarrow \lceil n \rceil & & \\
 1 & \xrightarrow{\lceil n \rceil} & \mathbf{N} & \longmapsto & \mathbf{N} & \longrightarrow & \mathbf{N} \longmapsto
 \end{array}
 \begin{cases} \{n-1\} & , n \geq 0 \\ \emptyset & , n = 0 \end{cases}$$

$$x \longmapsto x + 1$$

□

Thus:

- For $n = 1$: $(\zeta * \epsilon)(\lceil 1 \rceil) = 1$, $(\zeta * \delta^1)(\lceil 1 \rceil) = \{n-1\} \simeq 1$, $\varepsilon(\lceil 1 \rceil) = \emptyset$
- For $n = 0$: $(\zeta * \epsilon)(\lceil 0 \rceil) = 1$, $(\zeta * \delta^1)(\lceil 0 \rceil) = \emptyset$, $\varepsilon(\lceil 0 \rceil) = 1$.

Hence, for $n \leq 1$,

$$\zeta * \Phi_0 = \zeta * \delta^1 + \varepsilon.$$

Combining the cases $n \geq 2$ and $n \leq 1$, we obtain the objective-level Möbius principle for $\mathbf{N}$.

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